

AIML Capstone 2: Preserving Heritage / Tourism

Part 1: heritage structure classification (TensorFlow transfer learning). Part 2: Indonesia tourism EDA and collaborative filtering recommender.

1. Business context

Government agency monitors historical structures and promotes tourism via personalised recommendations.

2. Part 1 — Heritage classification

Dataset. 10 architectural element classes; train + unlabelled test set.

Exploration. Sample images per class, class distribution, OpenCV previews.

Model. MobileNetV2 (ImageNet weights), frozen conv layers, GAP → Dropout → Dense softmax head.

Training. Without augmentation first, then with augmentation; EarlyStopping, ModelCheckpoint, custom callback stopping at $\text{val_accuracy} \geq 0.92$.

Evaluation. Train/val accuracy curves; inference on Dataset_test images.

3. Part 2 — Tourism recommender

Datasets. 437 places, 10,000 ratings (9,921 after dedup), 300 users.

Cleaning. Imputed Time_Minutes median; removed 79 duplicate ratings.

EDA. User age/location profiles; category mix by city; Yogyakarta best for nature (57 Cagar Alam/Bahari spots).

Ratings. Yogyakarta highest mean place rating; Taman Hiburan most liked category (avg 3.12).

Recommender. Item-based CF: 437×300 rating matrix, cosine similarity + KNN; top-K places given current location (e.g. Monumen Nasional).

4. Tools

Part 1: TensorFlow/Keras, OpenCV (GPU/Colab). Part 2: pandas, sklearn, seaborn.

5. Conclusion

Transfer-learned classifier supports structure monitoring; CF recommender suggests similar destinations for tourists.